6A Solving Linear Equations

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. Solve each two-step linear equation:

(a) $5x - 9 = 26$ (b) $4x + 17 = 5$ (c) $19 - 3x = 34$ (d) $\frac{x}{6} - 4 = 3$

Q2. Solve each linear equation involving fractions or brackets:

(a) $\frac{3x - 5}{4} = 7$ (b) $\frac{2x + 9}{5} = -3$ (c) $4(2x - 3) = 28$ (d) $-3(x - 6) = 27$

Q3. Solve each equation with pronumerals on both sides:

(a) $8x - 5 = 3x + 30$ (b) $6x + 11 = 2x - 13$ (c) $14 - 2x = 5x - 28$ (d) $9x - 16 = 4x + 9$

Q4. Solve each multi-step linear equation:

(a) $3(2x + 5) = 2(x + 11) + 9$ (b) $5(x - 2) - 2(x - 4) = 16$ (c) $\frac{3x - 2}{4} = \frac{x + 3}{2}$ (d) $\frac{4x + 1}{3} = \frac{2x + 5}{2}$

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. A rectangular tennis court has a perimeter of $72\ \text{m}$. Its length is $3\ \text{m}$ more than twice its width w. Form and solve a linear equation to find the width w and length l.

Q6. Two kayak hire companies on the Yarra River charge different rates. RiverGlide charges a $\$30$ safety briefing fee plus $\$22$ per hour. PaddleMelb charges a $\$12$ briefing fee plus $\$28$ per hour. For how many hours h of hire are their total charges equal, and what is that charge?

Q7. The sum of four consecutive integers is 138. Let the smallest integer be n. Form a linear equation and find all four integers.

Q8. An adult ticket to Healesville Sanctuary costs $\$14$ more than a child ticket c. A family of 2 adults and 3 children pays a total of $\$158$. Form an equation in terms of c and find the price of a child ticket and an adult ticket.

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. Find the value of the constant m so that $x = 4$ is the solution to $3(2x - m) = 4x + 2$. Show your substitution and working.

Q10. Explain how you can tell without graphing whether the linear equation $4(x - 3) = 2(2x - 6)$ has one solution, no solution, or infinitely many solutions.